QCM Exam - Artificial Intelligence

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Instructions: There is only one good answer for each question. All questions are based on the course slides.

Each question is worth 1 point. There are no negative points.

First Name: ______________________ **Last Name:** ______________________

1. What is the main goal of Artificial Intelligence?
   A. To replace humans in all physical jobs.
   B. To design machines capable of performing tasks that usually require human intelligence.
   C. To build faster and more energy-efficient computers.
   D. To solve complex mathematical equations symbolically.

2. Alan Turing invented artificial intelligence through the decryption of Enigma. (True or False)
   A. True
   B. False

3. Which breakthrough launched the modern deep learning era?
   A. The Turing Test (1950)
   B. The Dartmouth workshop (1956)
   C. Expert systems (1980s)
   D. AlexNet / ImageNet (2012)

4. Which statement is TRUE?
   A. Machine Learning is broader than AI.
   B. AI is broader than ML.
   C. AI and ML are exact synonyms.
   D. Deep Learning is broader than AI.

5. Predicting whether an image contains a cat or a dog is an example of:
   A. Regression
   B. Clustering
   C. Classification
   D. Natural Language Processing

6. Which task belongs to unsupervised learning?
   - A. Regression
   - B. Classification
   - C. Clustering
   - D. Game AI
7. In linear regression $y = wx + b$, what do we aim to optimize?
   - A. x and y
   - B. w and b
   - C. y only
   - D. x only
8. What is the purpose of a loss function?
   - A. Measure model error.
   - B. Add non-linearity.
   - C. Initialize the parameters.
   - D. Normalize the dataset.
9. If the gradient of a weight is positive, what will happen to the value of that weight during gradient descent?
   - A. Decrease
   - B. Increase
   - C. Remain fixed
   - D. Explode
10. According to the course, what is one major limitation of AI?
    - A. It can only learn from unstructured text.
    - B. Data dependency (No data, no learning).
    - C. It perfectly generalizes to any unseen situation.
    - D. It operates without any computational cost.
11. A larger model will always perform better than a smaller model.
    - A. True
    - B. False
12. What happens if we stack two linear layers without an activation function?
    - A. It creates a highly non-linear model.
    - B. It is mathematically equivalent to a single linear layer.
    - C. It automatically normalizes the data.
    - D. It becomes a Convolutional Neural Network.
13. Why are convolutional layers better suited for images?
    - A. They flatten the image into a 1D vector instantly.
    - B. They keep spatial context through local filters.

C. They ignore color channels.

D. They require more parameters than a linear layer.

14. A batch size of 100 is used on a dataset of 1000 samples. How many gradient updates are performed in one epoch?

    A. 1

    B. 10

    C. 100

    D. 1000

15. In `nn.Linear(64, 256)`, what does 256 represent?

    A. Input features

    B. Output neurons

    C. Batch size

    D. Number of layers

16. What is the main role of a ReLU activation?

    A. Non linearity

    B. Normalize the output

    C. Add spatial context

    D. Decrease the loss

17. What does MaxPool2D keep from each local window?

    A. The average value

    B. The minimum value

    C. Select maximum value

    D. The sum of all values

18. What is the main purpose of Batch Normalization?

    A. Make the model smaller

    B. Add non-linearity

    C. Stabilize learning

    D. Perform classification

19. LayerNorm normalizes each feature across samples, while BatchNorm normalizes all features within a single sample.

    A. True

    B. False

20. How many trainable parameters does `nn.Linear(10, 5)` contain?

    A. 15

    B. 50

    C. 55

    D. 60

21. How many trainable parameters does this model contain?

```
import torch.nn as nn
model = nn.Sequential(
    nn.Linear(10, 20),
    nn.ReLU()
)
```

A. 200
B. 220
C. 240
D. 20

22. What is the Mean Squared Error (MSE) typically used for?

A. To add non-linearity to the model
B. As an activation function for classification
C. As a loss function for regression tasks
D. To normalize the dataset

23. Which of the following is an example of regression?

A. Predicting a continuous value like temperature
B. Identifying a cat or a dog in an image
C. Grouping customers into segments
D. Choosing the best move in a chess game

24. Grouping similar data points without predefined labels is known as:

A. Supervised learning
B. Unsupervised learning
C. Game AI
D. Reinforcement learning

25. What is the definition of an "Epoch" in neural network training?

A. A subset of the training data used for one gradient update
B. The initialization phase of the weights
C. The model has seen the entire training dataset once
D. A single forward pass through the network

26. Which of the following is a property of a good loss function?

A. Can be negative
B. Zero for perfect prediction
C. Unstable gradients
D. Non-differentiable

27. If the gradient is 0 during training, what happens to the parameters?

A. They increase
B. They decrease

C. They remain fixed

D. They are reset to zero

28. Does the initial state (random initialization) of the weights prevent a simple linear regression from converging?

    A. Yes, it will completely fail to converge.

    B. No, gradient descent will move towards the optimum regardless.

    C. Only if the learning rate is 0.

    D. Yes, it requires exact initialization.

29. In Deep learning models, what happens if the learning rate (η) is too large?

    A. Training is very slow but safe.

    B. There is a risk that the parameters overshoot the optimal values.

    C. The model perfectly fits the training data immediately.

    D. It automatically reduces the number of epochs.

30. Why do we often standardize (normalize) features before training?

    A. Because unbalanced scales cause unstable and unbalanced gradients.

    B. Because it increases the computational cost of the model.

    C. Because it acts as an activation function.

    D. Because it reduces the number of parameters in the model.

31. In modern language models, we often seek to:

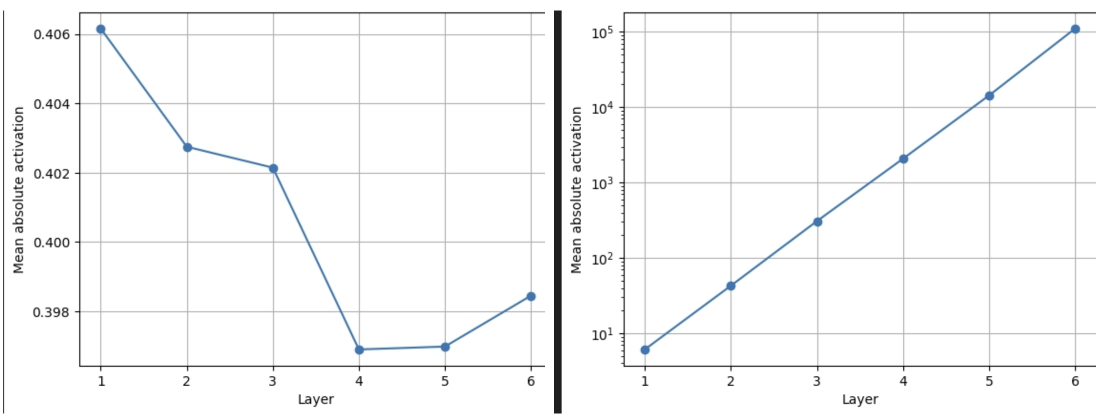
    A. Predict the next word knowing the context.

    B. Exploit the local spatial structure of an image.

    C. Group similar data points without predefined labels.

    D. Minimize the size of the convolution kernel.

32. What is the difference between the two plots?



    A. Normalized activation vs Exploding activation.

    B. High learning rate vs Low learning rate.

    C. Overfitting vs Underfitting.

    D. Linear layers vs Convolutional layers.

33. What is the shape of the weight tensor for a PyTorch `nn.Conv2d(in_channels=3, out_channels=16, kernel_size=5)` layer?
    A. `[16, 3, 5, 5]`
    B. `[3, 16, 5, 5]`
    C. `[16, 5, 5]`
    D. `[3, 5, 5]`
34. A standard scalar linear regression $\hat{y} = wx + b$ can be viewed as:
    A. A full convolutional layer.
    B. One artificial neuron without an activation function.
    C. A deep neural network.
    D. An attention layer.
35. In PyTorch, what is the shape of the weight matrix W for the layer `nn.Linear(64, 256)`?
    A. `[64, 256]`
    B. `[256, 64]`
    C. `[256]`
    D. `[64]`
36. What is the mathematical formula for the ReLU activation function?
    A. $\max(0, x)$
    B. $\min(0, x)$
    C. $1/(1 + e^{-x})$
    D. x^2
37. An image with RGB channels is mathematically represented as a:
    A. 1D vector
    B. 2D matrix
    C. 3D tensor
    D. 4D tensor
38. What does the `kernel_size` define in a Convolutional layer?
    A. The spatial size of the local filter window.
    B. The number of output channels.
    C. The amount of padding added to the borders.
    D. The size of the batch.
39. Mathematically performing a cognitive task is similar to:
    A. Approximating a complex mathematical function.
    B. Solving a simple linear equation.
    C. Memorizing a database perfectly.
    D. Generating random numbers.
40. Why is MaxPool used in Convolutional Neural Networks?

A. To add new learnable parameters.
B. To increase the spatial resolution of the image.
C. To reduce computation, memory usage, and improve robustness to small translations.
D. To replace the ReLU activation.

41. Which normalization method normalizes features within a single example (across its features)?
    A. BatchNorm
    B. LayerNorm
    C. MinMaxNorm
    D. Z-Score

42. Which architecture was introduced in 2017 and is based on attention mechanisms?
    A. Expert Systems
    B. ResNet
    C. Transformer
    D. AlexNet

43. How is a convolutional filter applied to an image?
    A. Only once on the entire image
    B. Repeatedly across different spatial locations
    C. Independently on each output class
    D. After the loss function is computed

44. How many parameters are in a Convolutional layer with `in_channels=2`, `out_channels=1`, and a `kernel_size=3x3`?
    A. 18
    B. 19
    C. 9
    D. 36

45. What is the main role of gradient descent?
    A. To find the minimum of the loss function by updating the model's parameters.
    B. To normalize the dataset before training.
    C. To calculate the accuracy of the model on the test set.
    D. To initialize the parameters with random values.

46. Which of the following is the equation of the standardization (where μ is the mean and σ is the standard deviation)?
    A. $z = \frac{x-\mu}{\sigma}$
    B. $z = \frac{x-x_{min}}{x_{max}-x_{min}}$
    C. $z = x - \mu$
    D. $z = \frac{x}{\sigma}$

47. What is the gradient descent update rule for a parameter W?
    A. $W \leftarrow W + \eta \frac{\partial \mathcal{L}}{\partial W}$
    B. $W \leftarrow W - \eta \frac{\partial \mathcal{L}}{\partial W}$
    C. $W \leftarrow W \times \frac{\partial \mathcal{L}}{\partial W}$
    D. $W \leftarrow \eta \frac{\partial \mathcal{L}}{\partial W}$

48. Consider the following PyTorch model. Is this model optimal for an image classification problem?

```
import torch.nn as nn

model = nn.Sequential(
    nn.Conv2d(3, 16, kernel_size=3),
    nn.BatchNorm2d(16),
    nn.Conv2d(16, 32, kernel_size=3),
    nn.BatchNorm2d(32),
    nn.Flatten(),
    nn.Linear(32 * 28 * 28, 10)
)
```

    A. No, it lacks an activation function.
    B. No, there is no point adding a linear layer after a convolution, it just increases the size of the model.
    C. No, because a CNN cannot be used for image classification without LayerNorm.
    D. Yes, this is a standard and optimal Convolutional Neural Network architecture.

49. Which of the following statements about linear layers vs CNNs on images is true based on the course?
    A. Linear layers always achieve higher accuracy than CNNs on images.
    B. CNNs obtain better accuracy with fewer parameters because they exploit the spatial structure.
    C. Linear layers preserve the 2D spatial context of pixels perfectly.
    D. CNNs cannot be combined with BatchNorm.